

Linear Matrix Inequalities for Robust Control

Qwen3.8-Flash-Next

A plant whose matrices depend on an unknown but bounded operator can be nominally stable yet fail somewhere in its uncertainty set. This note asks six questions about certifying such a family with linear matrix inequalities: what a certificate must quantify over, why no finite set of parameter values can decide it, what extra object makes the “for all” constraint affine, why one bound covers the whole family and whether it is conservative, how a state-feedback controller changes the problem class, and what changes when the uncertainty is structured or repeated. The running example is a linear-fractional plant with one norm-bounded uncertainty block.

October 01, 2026

Contents

1. Why does a nominal certificate say nothing about a family?	3
2. Why can no finite parameter sample decide the question?	4
3. What extra object makes the infinite constraint affine?	5
4. Why should one bound cover every value of the uncertainty?	6
5. What does the method produce, and where does a controller enter?	7
6. What changes when the uncertainty is structured or repeated?	8
7. Sources and further reading	9

1. Why does a nominal certificate say nothing about a family?

Question: A Lyapunov certificate is a single quadratic function of the state. What must hold for one such function to certify a whole family of plants rather than one member, and why does a P that satisfies the nominal Lyapunov inequality fail to certify the family even when it happens to certify some other member?

Running example. Consider the linear-fractional plant

$$\dot{x} = Ax + B_w w, \quad z = C_z x, \quad w = \Delta z, \quad \|\Delta\|_2 \leq 1, \quad (1)$$

with $x \in \mathbb{R}^n$, $w \in \mathbb{R}^m$, $z \in \mathbb{R}^p$, known matrices A, B_w, C_z , and an unknown constant operator $\Delta \in \mathbb{R}^{m \times p}$ of spectral norm at most one. Eliminating the loop $w = \Delta z$ gives the uncertain closed loop

$$\dot{x} = (A + B_w \Delta C_z)x = A(\Delta)x, \quad A(\Delta) \in \mathcal{A} := \{A + B_w \Delta C_z : \|\Delta\|_2 \leq 1\}. \quad (2)$$

Facts to reason with.

- A is Hurwitz; B_w and C_z are fixed and known.
- Δ is constant in time but its value is unknown; only the bound $\|\Delta\|_2 \leq 1$ is available.
- A quadratic Lyapunov candidate is $V(x) = x^T P x$ with $P = P^T \succ 0$, and along Equation 2 it has $\dot{V} = x^T (A(\Delta)^T P + P A(\Delta)) x$.
- The nominal plant corresponds to $\Delta = 0$ and its Lyapunov inequality is $A^T P + P A \prec 0$.
- A scalar instance: $A = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}$, $B_w = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $C_z = (0.8 \ 0.8)$, and a scalar $\delta \in [-1, 1]$. The nominal plant is stable, yet at $\delta = 1$ the closed loop $\begin{pmatrix} -0.2 & 0.8 \\ 0.8 & -1.2 \end{pmatrix}$ has determinant $-0.4 < 0$ and is unstable.

2. Why can no finite parameter sample decide the question?

Question: The robust condition must hold at every point of the uncertainty set. For each fixed Δ it is one matrix inequality, so it can be tested at any finite list of values. Why can no finite list ever stand in for the whole set, and where does the obstruction live: in the size of the set, in how the unknowns enter, or in both?

The robust condition. We want a common certificate P with

$$A(\Delta)^T P + P A(\Delta) \prec 0 \quad \text{for all } \|\Delta\|_2 \leq 1, \quad (3)$$

and expanding $A(\Delta)$ from Equation 2 gives

$$A(\Delta)^T P + P A(\Delta) = A^T P + P A + C_z^T \Delta^T B_w^T P + P B_w \Delta C_z. \quad (4)$$

Facts to reason with.

- For each fixed Δ , Equation 3 is a single matrix inequality affine in P .
- The admissible set $\Delta : \|\Delta\|_2 \leq 1$ is a continuum; for a matrix ball the extreme points are the entire boundary, not a finite set of corners.
- In Equation 4 the term $P B_w \Delta C_z$ multiplies the unknown certificate P by the unknown operator Δ .
- A grid or vertex check tests Equation 3 only at the listed values of Δ .

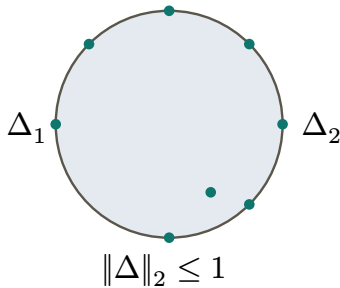


Figure 1: The admissible set $\|\Delta\|_2 \leq 1$ drawn as a disk, with several sample points marked. Inspect what the marked points determine about the set, and what they leave untested.

3. What extra object makes the infinite constraint affine?

Question: The obstruction is the cross term $C_z^T \Delta^T B_w^T P + P B_w \Delta C_z$, a sum $G + G^T$ of a product of the unknowns. What matrix generalization of the scalar inequality below bounds this term by an expression affine in the certificate, and what must that new object satisfy relative to Δ before the bound $\Delta^T \Delta \preceq I$ can be turned into a statement about Δ alone?

The goal. Replace the quantified family Equation 3 by a single matrix inequality in the certificate and one added variable, so that feasibility can be decided in one shot.

Facts to reason with.

- For scalars and any $\gamma > 0$,

$$2uv \leq \gamma u^2 + \gamma^{-1} v^2. \quad (5)$$

- The uncertain cross term in Equation 4 is $G + G^T$ with $G = P B_w \Delta C_z$, and $G + G^T$ is symmetric.
- The only information available about Δ is $\Delta^T \Delta \preceq I$.
- The new object must keep the final inequality affine in the unknowns.

4. Why should one bound cover every value of the uncertainty?

Question: The test you built bounds the uncertain cross term once, yet it must hold for every admissible Δ . Could that test reject a family that is in fact robustly stable? What single feature of the constraint $\Delta^T \Delta \preceq I$ decides whether the bound can lose anything, and what change to the uncertainty set would make it lose?

Setup. The bound replaces the quantified requirement Equation 3 by one matrix inequality in the certificate and the added object. The admissible set of the running example is the single quadratic constraint $\Delta^T \Delta \preceq I$.

Facts to reason with.

- A sufficient test can be conservative: it may fail even though some certificate exists.
- The bound was obtained from a quadratic inequality in Δ , so it is itself one quadratic form in the uncertainty.
- The uncertainty here is one full block, so its admissible set is one quadratic constraint.
- A test that is conservative loses certificates rather than gaining them.

5. What does the method produce, and where does a controller enter?

Question: With the added object in place, analysis becomes one matrix inequality in the certificate and that object. Now add a state-feedback controller $u = Kx$, so the closed loop contains B_2K next to P . A congruence by $Q = P^{-1}$ and the substitution $Y_c = KQ$ remove the controller from the products. Which terms does that substitution linearize, which terms stay coupled, and what class of problem does the result become?

Setup. Augment the plant Equation 1 with a control channel,

$$\dot{x} = Ax + B_2u + B_w w, \quad z = C_z x, \quad w = \Delta z, \quad \|\Delta\|_2 \leq 1, \quad (6)$$

and look for $u = Kx$ such that the closed loop $A + B_2K + B_w\Delta C_z$ satisfies the robust condition for every admissible Δ . Let $Y \succ 0$ be the added object; carrying it through the bound gives the condition

$$(A + B_2K)^T P + P(A + B_2K) + PB_w Y^{-1} B_w^T P + C_z^T Y C_z \prec 0. \quad (7)$$

Facts to reason with.

- Equation 7 is affine in P and affine in K , but contains the product PB_2K and the product $PB_w Y^{-1} B_w^T P$.
- Congruence by $Q = P^{-1}$ and the substitution $Y_c = KQ$ rewrite $Q(A + B_2K)^T$ as $QA^T + Y_c^T B_2^T$ and $(A + B_2K)Q$ as $AQ + B_2Y_c$.
- The added object still appears as Y and Y^{-1} , and the term $QC_z^T Y C_z Q$ contains Q twice.
- The uncertainty is still one full block.

6. What changes when the uncertainty is structured or repeated?

Question: So far the uncertainty is one full block. Replace it by two independent scalar blocks, $\Delta = \begin{pmatrix} \delta_1 I_{m_1} & 0 \\ 0 & \delta_2 I_{m_2} \end{pmatrix}$ with $|\delta_1| \leq 1$ and $|\delta_2| \leq 1$. What happens to the guarantee of the test, what must change about the added object, and does that change restore exactness?

Setup. Keep the plant Equation 6 and the condition, but let the uncertainty be block-diagonal with two independent blocks of sizes m_1 and m_2 ,

$$\Delta = \begin{pmatrix} \delta_1 I_{m_1} & 0 \\ 0 & \delta_2 I_{m_2} \end{pmatrix}, \quad |\delta_1| \leq 1, \quad |\delta_2| \leq 1. \quad (8)$$

Facts to reason with.

- The test so far used one added object, sized for the single block.
- The admissible set is now the intersection of two constraints, $\delta_1^2 \leq 1$ and $\delta_2^2 \leq 1$, that are not a single quadratic form.
- The added object must stay positive definite.
- The test must still be a single matrix inequality in the certificate and that object.

7. Sources and further reading

No external sources have been consulted yet; this note makes no external claims.